



BAULKHAM HILLS HIGH SCHOOL

2017 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks – 100

Exam consists of 10 pages.

This paper consists of TWO sections.

Section I – Page 2-4 (10 marks)

- Attempt Question 1-10
- Allow about **15 minutes** for this section

Section II – Pages 4-10 (90 marks)

- Attempt questions 11-16
- Allow about **2 hours and 45 minutes** for this section

Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer page for questions 1-10

1. Simplify $\frac{x^2-4}{x^3-8}$

- (A) $\frac{x-2}{x^2-2x+4}$
- (B) $\frac{1}{x+2}$
- (C) $\frac{x+2}{x^2+4x+4}$
- (D) $\frac{x+2}{x^2+2x+4}$

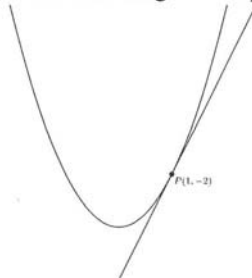
2. A yacht sailed directly from A to B on a bearing of 196°T . To sail from B back to A, the bearing should be:

- (A) 016°T
- (B) 074°T
- (C) 164°T
- (D) 196°T

3. Which of the following defines the domain of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$?

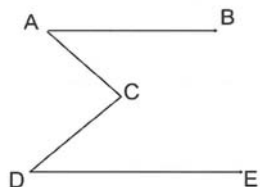
- (A) $x < -2$ or $x > 2$
- (B) $x < -2$
- (C) $x > 2$
- (D) $-2 < x < 2$

4. The diagram shows a concave up parabola with a tangent at the point $P(1, -2)$.



Which of the following could be the equation of the normal at P?

- (A) $x - 3y + 5 = 0$
 (B) $2x - 3y + 1 = 0$
 (C) $x + 3y + 5 = 0$
 (D) $x + 3y - 5 = 0$
5. In the diagram, AB and DE are parallel, $\angle BAC = 30^\circ$ and $\angle CDE = 55^\circ$.



What is the size of $\angle ACD$?

- (A) 45°
 (B) 65°
 (C) 85°
 (D) 95°
6. What is the derivative of $\frac{e^x}{e^x + 1}$?

- (A) $\ln(e^x + 1)$
 (B) $2\ln(e^x + 1)$
 (C) $\frac{2e^{2x} + e^x}{(e^x + 1)^2}$
 (D) $\frac{e^x}{(e^x + 1)^2}$

7. The quadratic equation $3x^2 - 5x + 1 = 0$ has roots α and β . Which of the following statements is true?

- (A) $2\alpha\beta = -\frac{4}{3}$
 (B) $\alpha^2 + \beta^2 = \frac{13}{9}$
 (C) $2\alpha + 2\beta = \frac{10}{3}$
 (D) $\alpha^2\beta^2 = \frac{2}{9}$

8. In $\triangle ABC$, $AB=5\text{m}$, $AC=8\text{m}$ and $\angle ACB = 40^\circ$. If $BC=d$ metres, which equation could be solved to find the length of BC ?

- (A) $d^2 = 5^2 + 8^2 - 2 \times 5 \times 8 \sin 40^\circ$
 (B) $d^2 = 5^2 + 8^2 - 2 \times 5 \times 8 \cos 40^\circ$
 (C) $5^2 = 8^2 + d^2 - 2d \times 8 \sin 40^\circ$
 (D) $5^2 = 8^2 + d^2 - 2d \times 8 \cos 40^\circ$

9. The values of x for which the geometric series $2 + 4x + 8x^2 + \dots$ has a limiting sum are:

- (A) $x < \frac{1}{2}$
 (B) $x \leq \frac{1}{2}$
 (C) $-\frac{1}{2} < x < \frac{1}{2}$
 (D) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

10. What is the number of solutions for the equation $x = 10 \sin x$?

- (A) 4
 (B) 5
 (C) 6
 (D) 7

END OF SECTION I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question on the appropriate page in the writing booklet.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on the appropriate page in the answer booklet

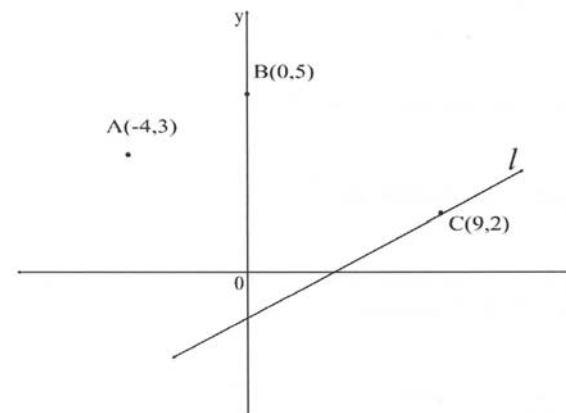
- (a) Factorise $2x^2 + x - 28$ 2
- (b) Solve $|x - 1| = 2x - 3$ 2
- (c) Find $\int \sqrt{3x - 4} \, dx$ 2
- (d) If $\tan \theta = \frac{5}{8}$ and $\sin \theta < 0$ find the exact value of $\operatorname{cosec} \theta$. 2
- (e) The points $A(4, 12)$, $B(8, 6)$ and $C(-4, k)$ are collinear. Find the value of k . 2
- (f) The quadratic equation $2x^2 - (4p + 1)x + 2p^2 - 1 = 0$ has two roots which are equal in magnitude but opposite in sign. Find the value of p . 2
- (g) Find the equation of the curve $y = f(x)$ given that $y = 5x - 7$ is a tangent to this curve, and its gradient function is $f'(x) = 4x - 3$. 3

Question 12 (15 marks) Start on the appropriate page in the answer booklet

(a) Differentiate with respect to x :

- (i) $(x^2 + 4)^5$ 2
- (ii) $e^x \log_e x$ 2
- (iii) $\frac{\sin x}{x-1}$ 2

(b) The diagram shows the points $A(-4, 3)$, $B(0, 5)$ and $C(9, 2)$, and the line l which passes through C and is parallel to AB .



Copy the diagram into your answer booklet.

- (i) Find the length AB . 1
- (ii) Show that the equation of line l is $x - 2y - 5 = 0$ 2
- (iii) Find the coordinates of point D , where line l meets the x -axis. 1
- (iv) Prove that $ABCD$ is a parallelogram. 2
- (v) Find the perpendicular distance from point B to line l . 2
- (vi) Hence or otherwise, find the area of parallelogram $ABCD$ 1

Question 13 (15 marks) Start on the appropriate page in the answer booklet

(a) Evaluate $\int_1^3 \frac{5x}{x^2 + 1} dx$

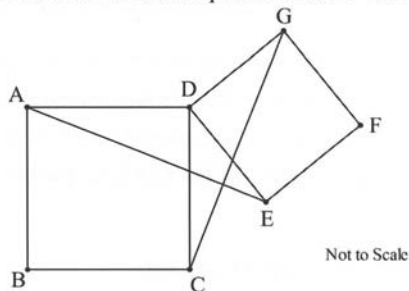
2

(b) For which values of k does the equation $3x^2 + kx + 5 = 0$ have no real solutions?

3

(c) In the diagram below, $ABCD$ and $DEFG$ are two squares. Prove $AE = CG$.

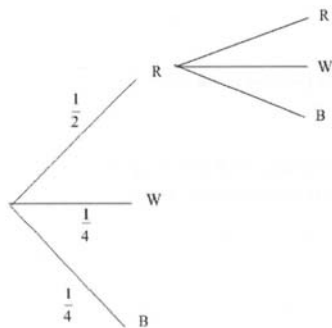
3



(d) A jar contains 2 red, 1 black and 1 white marble only. Ken randomly selects two marbles in succession (without replacement), and places them in a bag.

(i) Copy and complete the probability tree to show all possible outcomes.

2



(ii) Find the probability that both of the selected marbles are red.

1

(ii) If one of the marbles falls out of the bag and we see that it is white, what is the probability that the marble still in the bag is red?

1

(c) Evaluate $\sum_{n=1}^{20} (3^n + 2n - 1)$

3

Question 14 (15 marks) Start on the appropriate page in the answer booklet

(a) Given $\log_3 2 = p$ and $\log_3 5 = q$, find in terms of p and q :

(i) $\log_3 12.5$

1

(ii) $\log_5 2$

1

(b) The velocity of a particle is given by $v = 1 - 2 \cos t$ for $0 \leq t \leq 2\pi$, where v is measured in metres per second and t is measured in seconds.

(i) At what times during this period is the particle at rest?

2

(ii) What is the maximum velocity of the particle during this period?

2

(iii) Sketch the graph of v as a function of t for $0 \leq t \leq 2\pi$

2

(iv) Calculate the total distance travelled by the particle between $t = 0$ and $t = \pi$

3

(c) For the parabola $x^2 - 10x - 16y - 7 = 0$, find :

(i) the coordinates of the vertex.

2

(ii) the equation of the directrix.

1

(iii) the coordinates of the focus.

1

Question 15 (15 marks) Start on the appropriate page in the answer booklet

(a) Radium is a radioactive substance which decays over time, with a half-life of 1600 years. (Note: the half-life of a substance is the time taken for it to decay to half its mass).

The mass M remaining at time t years is given by:

$$M = M_0 e^{-kt}$$

where M_0 and k are constants.

(i) Evaluate k , expressing your answer correct to 3 significant figures.

2

(ii) If a piece of radium in a laboratory has a mass of 200 grams, what would have been its mass 1000 years ago? Answer correct to the nearest gram.

2

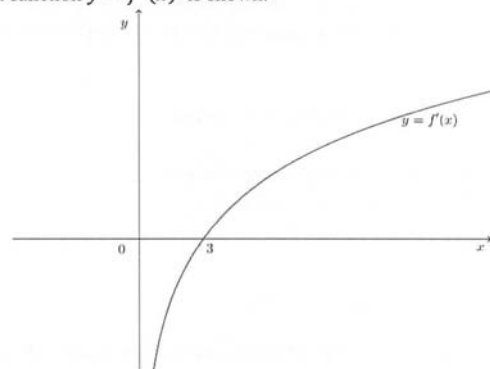
(b) Prove that $(1 - \cos \theta)(1 + \sec \theta) = \sin \theta \tan \theta$

2

Question 15 continues on the next page.

Question 15 (Continued)

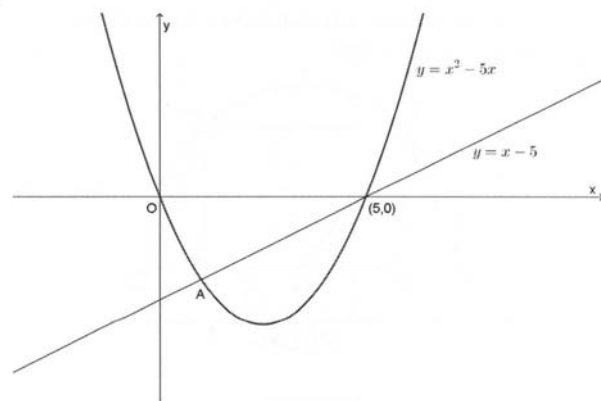
- (c) A sketch of a gradient function $y = f'(x)$ is shown.



Sketch the curve $y = f(x)$, given $f(3) = 0$.

3

- (d) The graphs of $y = x - 5$ and $y = x^2 - 5x$ intersect at the points (5,0) and A, as shown in the diagram.



- (i) Show that A has coordinates (1,-4).
 (ii) Find the area between the line $y = x - 5$ and the curve $y = x^2 - 5x$.
 (e) A sector of a circle is enclosed by two radii of length 10cm and an arc of length 14cm. Find the area of the sector.

1

3

2

Question 16 (15 marks) Start on the appropriate page in the answer booklet

- (a) Karina borrows \$80 000 to start a business. She plans to repay the loan in equal monthly instalments of \$M at 0.5% per month reducible interest. Interest is calculated and charged just before each repayment.

Let A_n = amount owing after n repayments (in dollars)

- (i) Show that the amount owing after 3 repayments is
 $A_3 = 80000(1.005)^3 - M(1.005^2 + 1.005 + 1)$

2

- (ii) Show that

$$A_n = 1.005^n(80000 - 200M) + 200M$$

2

- (iii) If she wishes to repay the loan in 10 years, calculate the amount of each instalment.

2

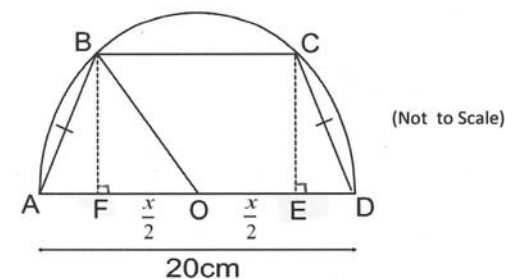
- (iv) If she can only repay \$700 per month, how long will it take her to repay the loan? Answer correct to the nearest year.

2

- (b) Solve $\sin(x + \frac{\pi}{6}) = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$

2

- (c) ABCD is a trapezium inscribed in a semicircle of diameter 20cm, as shown. $AB = CD$ and O is the centre of the semicircle.



- (i) If $EO = OF = \frac{x}{2}$, show that $BF = \frac{1}{2}\sqrt{400 - x^2}$

1

- (ii) Show that the area $A \text{ cm}^2$ of the trapezium ABCD is given by $A = \frac{1}{4}(x + 20)\sqrt{400 - x^2}$

1

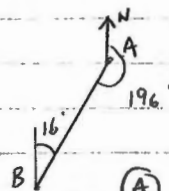
- (iii) Hence find the length of BC such that the area of the trapezium ABCD is a maximum

3

End of exam

1. $\frac{(x-2)(x+2)}{(x-2)(x^2+2x+4)}$

(D)



(A)

3. $4 - x^2 > 0$

$$4 > x^2$$

$$x^2 < 4$$

$$-2 < x < 2 \quad (D)$$

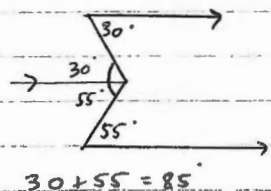
4. $m_{\text{tang.}} > 0$ $m_{\text{Norm.}} < 0$

C or D

but must pass through (4, 2)

Test C: $1 - 6 + 5 = 0$

Test D: $1 - 6 - 5 \neq 0$ $\therefore (C)$



$$30 + 55 = 85$$

(C)

6. $\frac{d}{dx} \left(\frac{e^x}{e^x + 1} \right)$

$$= \frac{(e^x + 1)e^x - e^x \cdot e^x}{(e^x + 1)^2}$$

$$= \frac{e^{2x} + e^x - e^{2x}}{(e^x + 1)^2}$$

(D)

7. $\alpha + \beta = \frac{5}{3}$, $\alpha\beta = -\frac{1}{3}$

From A, $\alpha\beta = -\frac{1}{3}$ NO.

From B, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \frac{25}{9} - \frac{2}{3}$$

$$= \frac{19}{9} \neq \frac{13}{9}$$

From C, $2\alpha + 2\beta = 2(\alpha + \beta)$

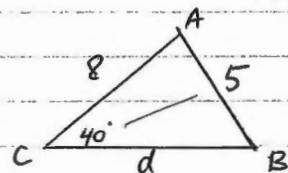
$$= 2\left(\frac{5}{3}\right)$$

$$= \frac{10}{3} \quad \underline{y_{20}}$$

From D, $\alpha^2\beta^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$

 $\therefore (C)$

8.



$$5^2 = 8^2 + d^2 - 2 \times 8 \times d \cos 40^\circ$$

(D)

9. $r = 2x$ $|k| < 1$

$$-1 < 2x < 1$$

$$-\frac{1}{2} < x < \frac{1}{2} \quad (C)$$

10. 7 (by graphing) (D)

Section II.

Question 11

a) $(2x-7)(x+4)$

2 - correct answer

[1 - signs incorrect eg $(2x+7)(x-4)$]

b) $|x-1| = 2x-3$

$$x-1 = 2x-3$$

$$2 = x$$

(Test: OK)

$$\text{or } -x+1 = 2x-3$$

$$-2 = 3x$$

$$x = -\frac{2}{3}$$

(Test: fails) RHS = $-\frac{4}{3} - 3 < 0$

$$\therefore x = 2 \text{ only}$$

2 - correct solution/testing.

[1 - leaving both]

c) $\int \sqrt{3x-4} \cdot dx$

$$= \frac{(3x-4)^{3/2}}{3/2 \times 3} + c$$

$$= \frac{2}{9} (\sqrt{3x-4})^3 + c \text{ or } \frac{2}{9} (3x-4)^{3/2} + c \quad 2 - \text{correct answer.}$$

d) $\sqrt{89}$ $\cos \theta = -\frac{\sqrt{89}}{6}$

2 - correct answer

[1 - finding $\sqrt{89}$]

e) $m_{AB} = m_{BC}$ $\frac{6-12}{8-4} = \frac{k-6}{-4-8}$

[1 - equating gradients]

$$\frac{-6}{4} = \frac{k-6}{-12}$$

$$k-6 = 18 \quad \therefore k = 24 \quad 2 - \text{correct solution}$$

f) Let roots = $\alpha, -\alpha$ $4(-\alpha) = \frac{4p+1}{2}$

$$0 = \frac{4p+1}{2}$$

$$p = -\frac{1}{4}$$

[1 - writing $\frac{4p+1}{2}$ for sum]

2 - correct solution.

g) $f'(x) = 4x - 3 = 5$, at the tangent
 $4x = 8$
 $x = 2, y = 5(2) - 7 = 3$ 1 - finding $x=2$
 $\therefore$ Pt. of contact is $(2, 3)$
 $f(x) = 2x^2 - 3x + c$ (by integration) 1 - finding primitive
 $3 = 2(2)^2 - 3(2) + c$
 $3 = 8 - 6 + c \therefore c = 1$
 $\therefore f(x) = 2x^2 - 3x + 1$ 1 - answer.

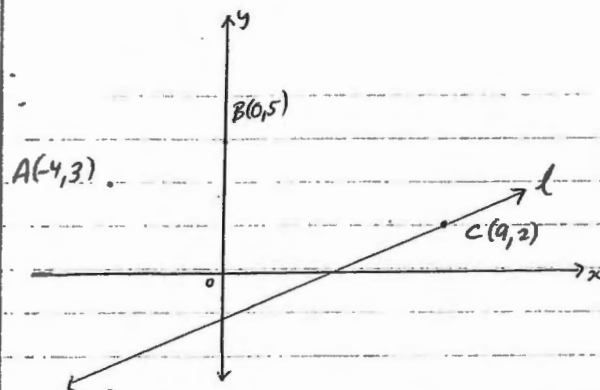
Question 12.

i) $\frac{d}{dx} (x^2 + 4)^5 = 5(x^2 + 4)^4 \cdot 2x$
 $= 10x(x^2 + 4)^4$

ii) $\frac{d}{dx} (e^x \cdot \log_e x) = uv' + vu'$ $\begin{cases} u = e^x & u' = e^x \\ v = \log_e x & v' = \frac{1}{x} \end{cases}$
 $= e^x \cdot \frac{1}{x} + \log_e x \cdot e^x$
 $= e^x \left(\frac{1}{x} + \log_e x \right)$ 2 - correct soln
 [1 - attempt prod. rule]

iii) $\frac{d}{dx} \left(\frac{\sin x}{x-1} \right) = \frac{vu' - uv'}{v^2}$ $\begin{cases} u = \sin x & u' = \cos x \\ v = x-1 & v' = 1 \end{cases}$
 $= \frac{(x-1) \cdot \cos x - \sin x \cdot 1}{(x-1)^2}$
 $= \frac{x \cos x - \cos x - \sin x}{(x-1)^2}$ 2 - correct soln.
 [1 - attempt quot. rule]

b)



(i) $AB = \sqrt{(5-3)^2 + (0+4)^2}$
 $= \sqrt{20}$ or $2\sqrt{5}$ units.

(ii) $m_{AB} = \frac{5-3}{0+4} = \frac{1}{2}$

l has $m = \frac{1}{2}$ and through $C(9, 2)$

$y - 2 = \frac{1}{2}(x - 9)$

$2y - 4 = x - 9$

$x - 2y - 5 = 0$

(iii) At D: $y = 0$ $x - 2(0) - 5 = 0$, $x = 5$

$\therefore D(5, 0)$

(iv) $AB \parallel CD$ (given)

$CD = \sqrt{(9-5)^2 + (2-0)^2} = \sqrt{20} = AB$

$\therefore ABCD$ is a parallelogram, since a pair of sides are equal and parallel.

v) $B(0, 5)$
 $l: x - 2y - 5 = 0$

$d = \frac{|Ax_1 + By_1 + c|}{\sqrt{A^2 + B^2}}$

$= \frac{|0 - 2(5) - 5|}{\sqrt{1^2 + (-2)^2}}$

$= \frac{15}{\sqrt{5}} = 3\sqrt{5}$

vi) Area = bh
 $= 2\sqrt{5} \times \frac{15}{\sqrt{5}}$
 $= 30 \text{ units}^2$

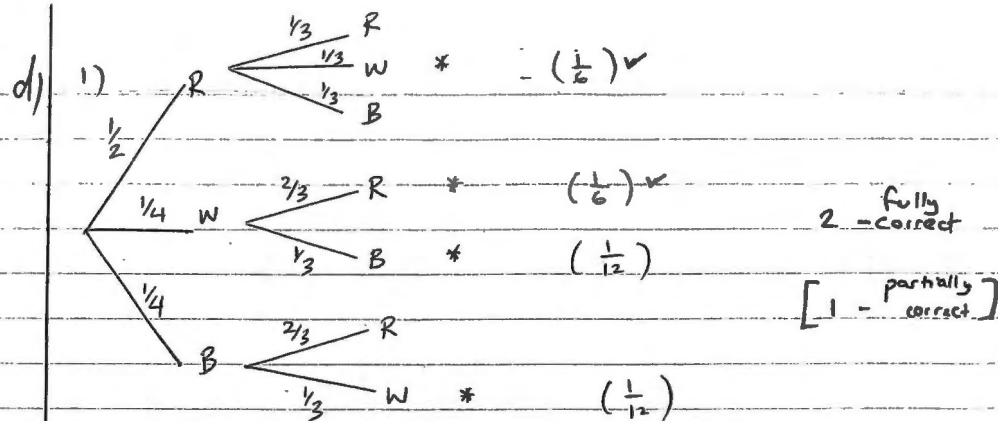


Question 13.

a) $\int_1^3 \frac{5x}{x^2+1} dx$
 $= \frac{5}{2} \int_1^3 \frac{2x}{x^2+1} dx$
 $= \frac{5}{2} [\ln(x^2+1)]_1^3$ ← correct primitive
 $= \frac{5}{2} (\ln 10 - \ln 2)$ ← correct substitution into found primitive.
 or $\frac{5}{2} \ln 5$.

b) $\Delta = b^2 - 4ac$
 $= k^2 - 4 \cdot 3 \cdot 5$ ← 1
 $= k^2 - 60 < 0$ ← if no real solutions
 $-\sqrt{60} < k < \sqrt{60}$ ← 1

c) In $\triangle ADE, \triangle DCG$:
 $AD = DC$ (sides of square)
 $\angle ADC = \angle GDE$ (Ls of square)
 $\angle ADE = \angle ADC + \angle CDE$
 $\angle GDC = \angle GDE + \angle CDE$
 $\therefore \angle ADE = \angle GDC$ (adding equal Ls)
 $DE = DG$ (sides of square)
 $\therefore \triangle ADE \equiv \triangle DCG$ (SAS) ← 1
 $\therefore AE = CG$ (matching sides of congruent Δ s equal) ← 1



ii) $P(RR) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ ← 1

iii) $P(1 \text{ red, given 1 white}) = \frac{\frac{1}{3}}{\frac{1}{2}}$ ← outcomes labelled ✓
 $= \frac{2}{3}$ ← outcomes labelled * ← 1

e) $\sum_{n=1}^{20} (3^n + 2n - 1)$

$= \sum_{n=1}^{20} 3^n + (2n - 1)$

$= (3 + 9 + 27 + \dots) + (1 + 3 + 5 + 7 + \dots + 39)$ ← 1 split + recognize AP, GP

$= \frac{a(r^n - 1)}{r - 1} + \frac{n}{2}(a + L)$
 ← sum of geom. series ← sum of arith. series

$= \frac{3(3^{20} - 1)}{3 - 1} + \frac{20}{2}(1 + 39)$ ← 1 sum of GP
 ← 1 sum of AP

$= 5230176600 + 400$

$= 5230177000$

Question 14

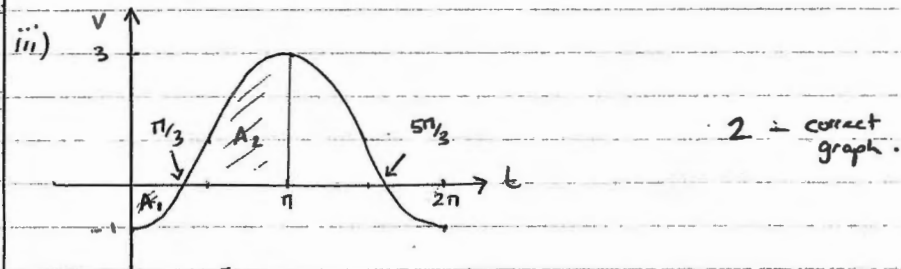
a) i) $\log_3 12.5 = \log_3 \frac{5^2}{2}$
 $= 2\log_3 5 - \log_3 2$

$= 2q - p.$ $\leftarrow 1$

ii) $\log_5 2 = \frac{\log_3 2}{\log_3 5} = \frac{p}{q}$ $\leftarrow 1$

b) i) $v = 0$ when $1 - 2\cos t = 0$
 $\cos t = \frac{1}{2}$ $\leftarrow 1$
 $t = \frac{\pi}{3}, \frac{5\pi}{3} \text{ sec}$ $\leftarrow 1$

ii) $\text{Max } v = 1 - 2(-1) = 3 \text{ m/s}$ $\leftarrow 1$



iv) $A_1 = \left| \int_0^{\pi/3} 1 - 2\cos t \cdot dt \right|$
 $= \left| \left[t - 2\sin t \right]_0^{\pi/3} \right|$
 $= \left| \frac{\pi}{3} - 2\left(\frac{\sqrt{3}}{2}\right) - (0 - 0) \right|$ $\leftarrow 1$

$= \sqrt{3} - \frac{\pi}{3}$

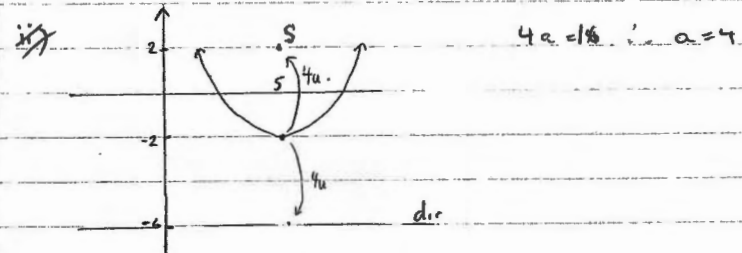
$A_2 = \int_{\pi/3}^{\pi} 1 - 2\cos t \cdot dt$

$= \left[t - 2\sin t \right]_{\pi/3}^{\pi}$

$= \pi - 2(0) - \left(\frac{\pi}{3} - 2\left(\frac{\sqrt{3}}{2}\right) \right)$

$\therefore \text{Distance travelled} = \sqrt{3} - \frac{\pi}{3} + \frac{2\pi}{3} + \sqrt{3}$
 $= 2\sqrt{3} + \frac{\pi}{3} \text{ cm}$ $\leftarrow 1$

c) i) $x^2 - 10x + 25 = 16y + 7 + 25$ $\leftarrow 1$ (completing the sq.)
 $(x-5)^2 = 16(y+2)$
 vertex $(5, -2)$. $\leftarrow 1$



ii) Directrix $y = -6$ $\leftarrow 1$

iii) Focus $(5, 2)$. $\leftarrow 1$

Question 15.

a) i) When $t = 1600$, $m = \frac{1}{2}m_0$

$\frac{1}{2}m_0 = m_0 e^{-1600k}$ $\leftarrow 1$

$-1600k = \ln \frac{1}{2}$

$k = 0.000433$ $\leftarrow 1$ (with correct rounding)

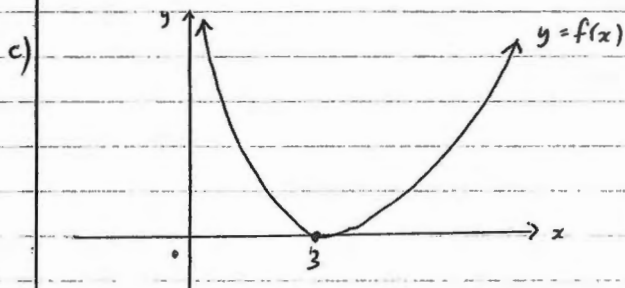
ii) When $t = 1000$, $m = 200$:

$200 = m_0 e^{-1000k}$ $\leftarrow 1$

$200 = m_0 \times 0.6484$

$m_0 = 308 \text{ g}$ $\leftarrow 1$

$$\begin{aligned}
 \text{b) LHS} &= (1 - \cos \theta)(1 + \sec \theta) \\
 &= 1 + \sec \theta - \cos \theta - \cos \theta \sec \theta \\
 &= 1 + \sec \theta - \cos \theta - \cancel{\cos \theta} \cdot \frac{1}{\cancel{\cos \theta}} \\
 &= \sec \theta - \cos \theta \quad \leftarrow 1 \\
 &= \frac{1 - \cos^2 \theta}{\cos \theta} \\
 &= \frac{\sin^2 \theta}{\cos \theta} \\
 &= \frac{\sin \theta}{\cos \theta} \cdot \sin \theta \quad \leftarrow 1 \\
 &= \tan \theta \cdot \sin \theta \\
 &= \text{RHS}
 \end{aligned}$$

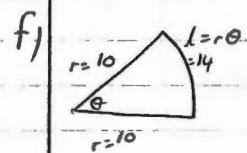


| - through (3,0)
 | - conc. up
 | - asymp.

d) i) At A : $x^2 - 5x = x - 5$
 $x^2 - 6x + 5 = 0$
 $(x-5)(x-1) = 0$
 ~~$x=5$~~ or $x=1$ $\leftarrow 1$
 $\uparrow$
 this is the x -int
 $y = 1 - 5 = -4$
 $\therefore A(1, -4)$

ii) $A = \int_1^5 (x-5) - (x^2-5x) dx$
 $= \int_1^5 -x^2 + 6x - 5 dx \quad \leftarrow 1$

$$\begin{aligned}
 &= \left[-\frac{x^3}{3} + 3x^2 - 5x \right]_1^5 \quad \leftarrow 1 \\
 &= \left(-\frac{125}{3} + 75 - 25 \right) - \left(-\frac{1}{3} + 3 - 5 \right) \\
 &= 10\frac{2}{3} \text{ units}^2 \quad \leftarrow 1
 \end{aligned}$$



$$\begin{aligned}
 A &= \frac{1}{2} r^2 \theta \\
 &= \frac{1}{2} r(l, \theta) \quad \leftarrow 1 \\
 &= \frac{1}{2} \times 10 \times 14 \\
 &= 70 \text{ cm}^2 \quad \leftarrow 1
 \end{aligned}$$

OR $l = r\theta \therefore 10\theta = 14$
 $\theta = 1.4 \quad \leftarrow 1$

Then $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 10^2 \times 1.4$
 $= 70 \text{ cm}^2 \quad \leftarrow 1$

v) If $M=700$ and $A_n=0$:

$$(80\,000 - 200 \times 700) \times 1.005^n + 200 \times 700 = 0 \quad \leftarrow$$

$$1.005^n = \frac{-140\,000}{-60\,000} = \frac{7}{3}$$

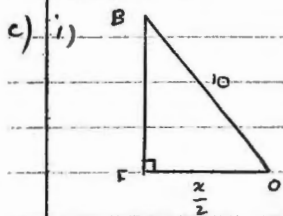
$$n \ln 1.005 = \ln \frac{7}{3}$$

$$n = \frac{\ln \frac{7}{3}}{\ln 1.005} = 170 \text{ months} \quad \leftarrow$$

) $\sin\left(x + \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ $\left(0 \leq x \leq 2\pi\right)$
 $\left(\frac{\pi}{6} \leq x + \frac{\pi}{6} \leq \frac{13\pi}{6}\right)$

$$x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{2\pi}{3} \quad \leftarrow$$

$$x = \frac{\pi}{6}, \frac{\pi}{2} \quad \leftarrow$$



$$BF^2 = 10^2 - \left(\frac{x}{2}\right)^2$$

$$= 100 - \frac{x^2}{4}$$

$$= \frac{400 - x^2}{4} \quad \leftarrow$$

$$BF = \sqrt{\frac{400 - x^2}{4}} = \frac{1}{2} \sqrt{400 - x^2}$$

ii) Area = $\frac{1}{2}(a+b)h$

$$\begin{cases} a=x \\ b=20 \\ h=BF \end{cases}$$

$$= \frac{1}{2}(x+20) \cdot \frac{1}{2} \sqrt{400-x^2} \quad \leftarrow$$

$$= \frac{1}{4}(x+20) \sqrt{400-x^2}$$

iii) For max area, $A' = 0$ and $A'' < 0$

$$A' = \frac{1}{4} \left((x+20) \cdot \frac{-x}{\sqrt{400-x^2}} + \sqrt{400-x^2} \cdot 1 \right)$$

= 0 when:

$$\frac{x(x+20)}{\sqrt{400-x^2}} = \sqrt{400-x^2}$$

$$x^2 + 20x = 400 - x^2 \quad \leftarrow$$

$$2x^2 + 20x - 400 = 0$$

$$x^2 + 10x - 200 = 0$$

$$(x+20)(x-10) = 0$$

$$x = -20 \quad \text{or} \quad x = 10 \quad \leftarrow$$

A'' is rather 'messy', so will use A' to test

x	9	10	11
A'	$\frac{1}{4}(324)$	0	$\frac{1}{4}(-37)$
	(+, 0, -)		



$\therefore$ Max area when $x=10$

Question 16

a) i) $A_1 = 80000(1.005) - M$

$$A_2 = A_1 \times 1.005 - M$$

$$= 80000(1.005)^2 - M(1.005) - M \quad \leftarrow 1$$

$$= 80000(1.005)^2 - M(1.005 + 1)$$

$$A_3 = A_2 \times 1.005 - M$$

$$= 80000(1.005)^3 - M(1.005^2 + 1.005) - M \quad \leftarrow 1$$

$$= 80000(1.005)^3 - M(1.005^2 + 1.005 + 1)$$

ii) Continuing the pattern

$$A_n = 80000(1.005)^n - M(1.005^{n-1} + 1.005^{n-2} + \dots + 1.005 + 1)$$

$$= 80000(1.005)^n - M \times \frac{a(r^n - 1)}{r - 1} \quad \begin{array}{l} \text{where} \\ a = 1 \\ r = 1.005 \\ n \text{ terms} \end{array}$$

$$= 80000(1.005)^n - M \times \frac{1(1.005^n - 1)}{0.005} \quad \leftarrow 1$$

$$= 80000(1.005)^n - M \times 200(1.005^n - 1)$$

$$= 80000(1.005)^n - 200M(1.005)^n + 200M \quad \leftarrow 1$$

$$= (80000 - 200M) \times 1.005^n + 200M$$

iii) If $n = 120$ and $A_{120} = 0$:

$$(80000 - 200M) \times 1.005^{120} + 200M = 0$$

$$80000(1.005)^{120} - (200M) \times 1.005^{120} + 200M = 0$$

$$80000(1.005)^{120} = 200M(1.005^{120} - 1) \quad \leftarrow 1$$

$$145561.74 = 200M \times 0.8194$$

$$M = \$ 888.16 \quad \leftarrow 1$$